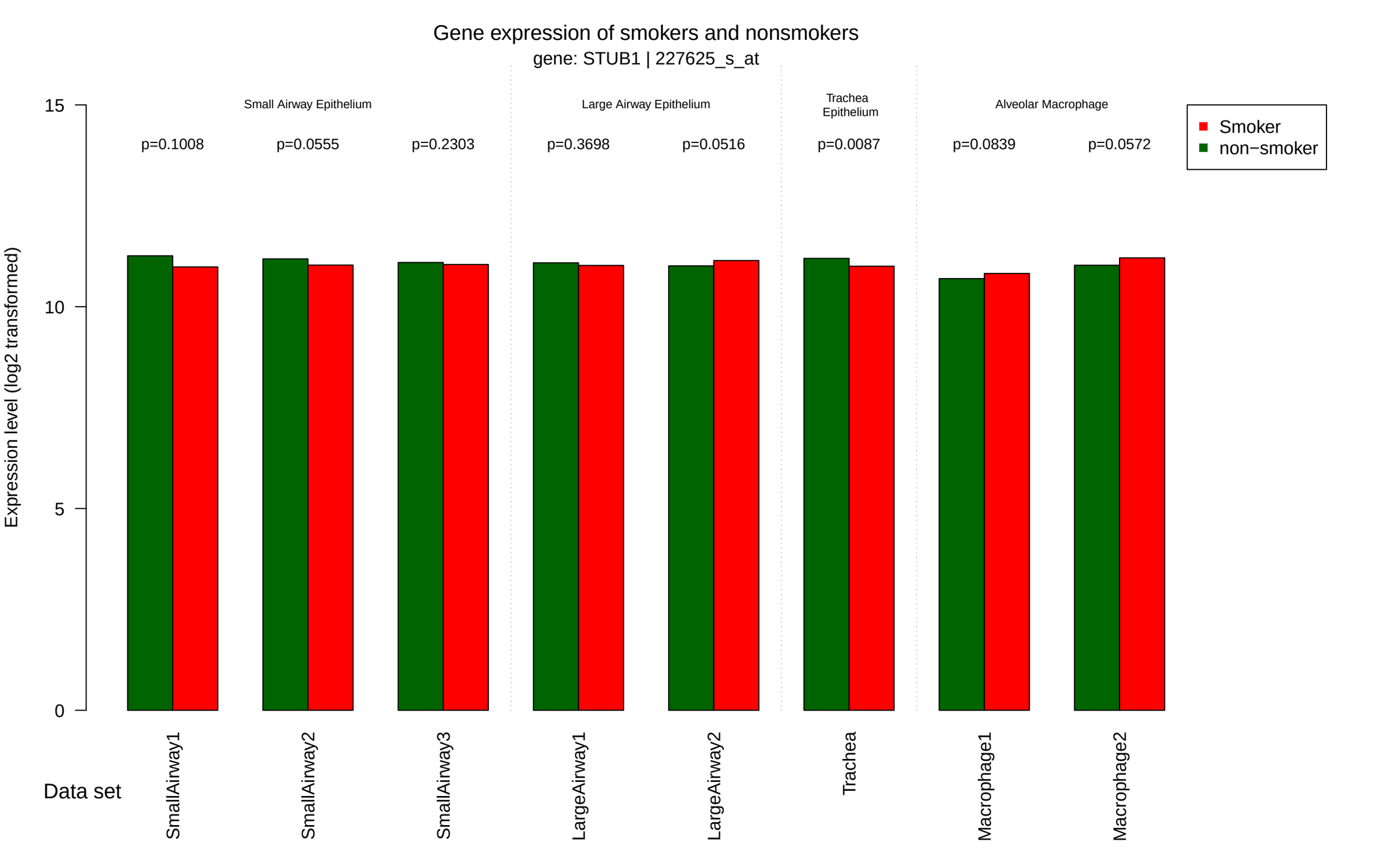
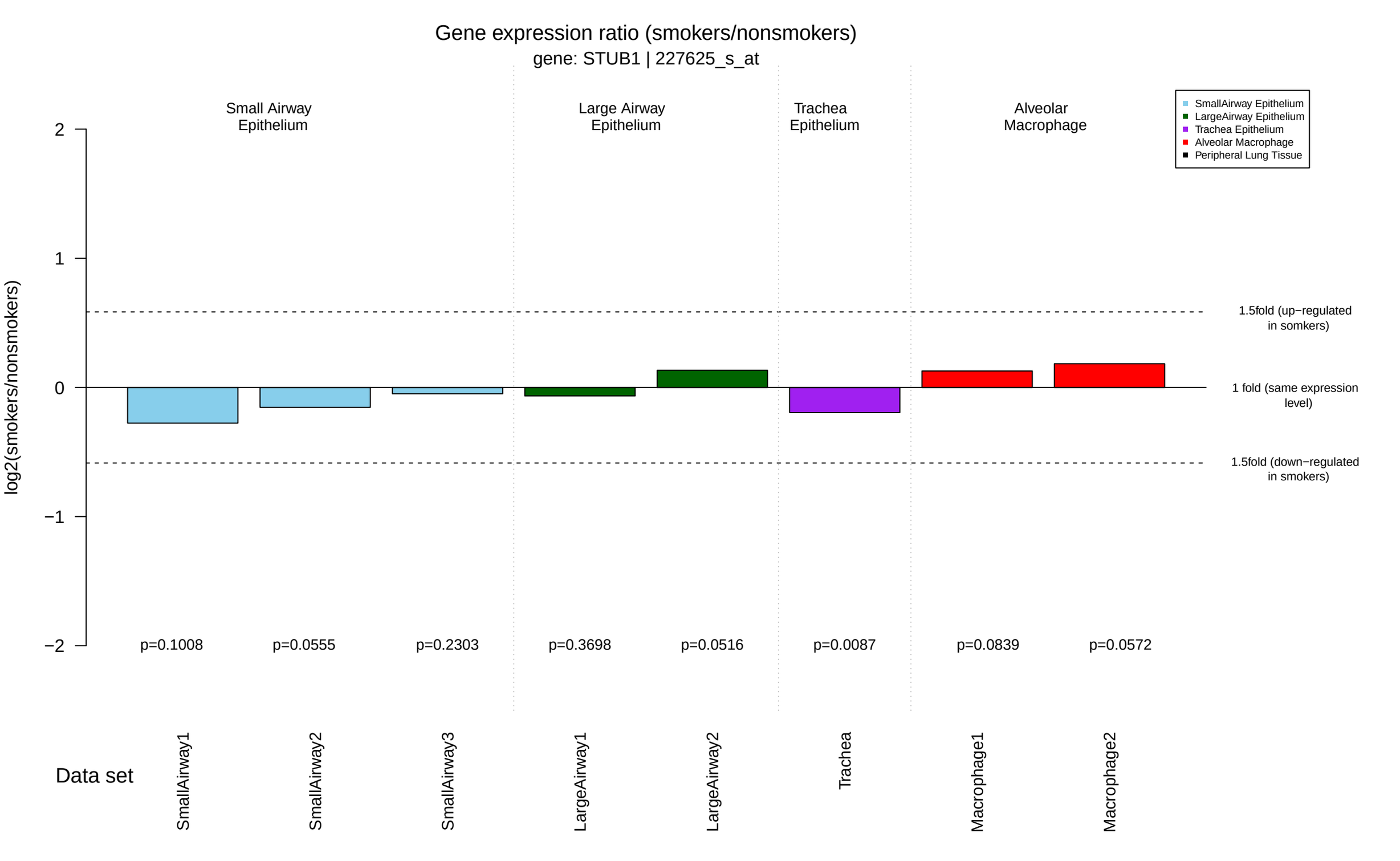
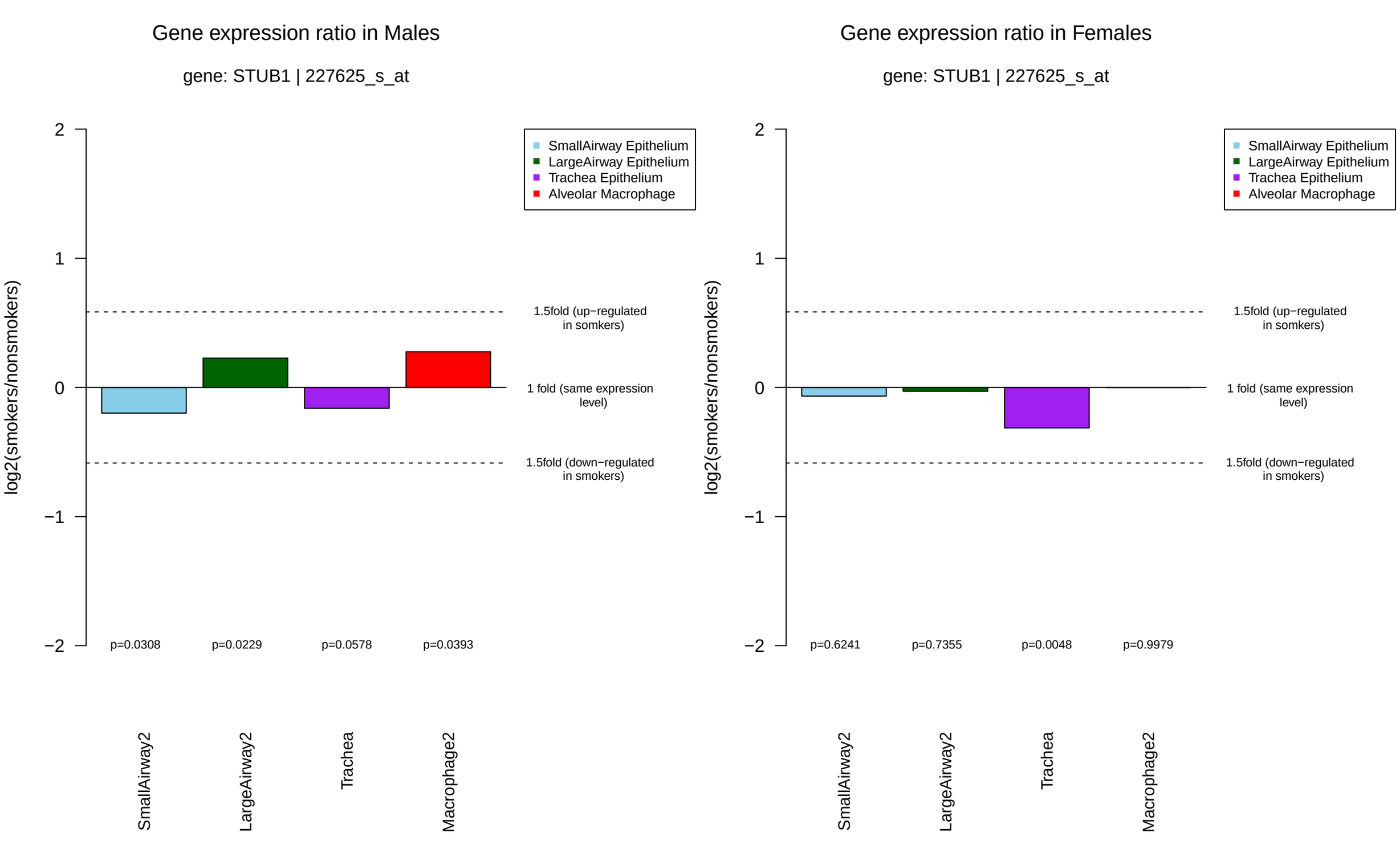
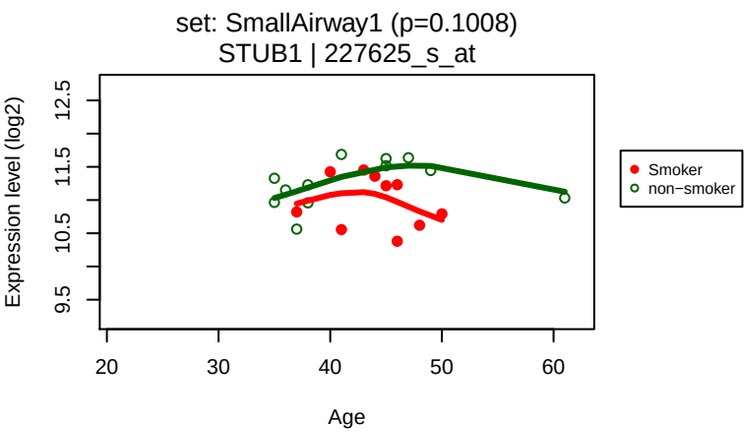
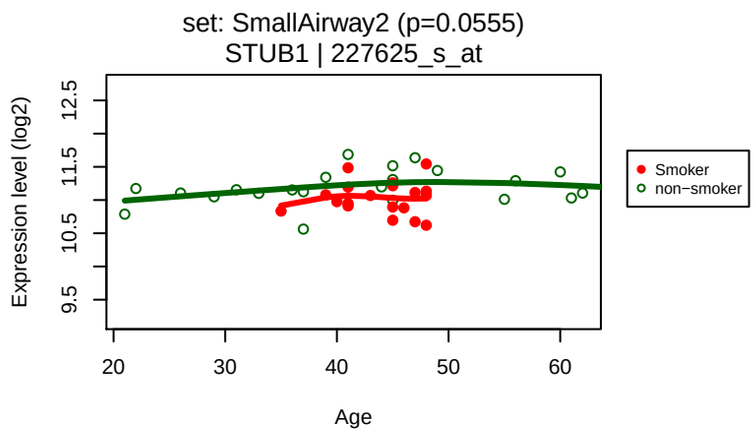
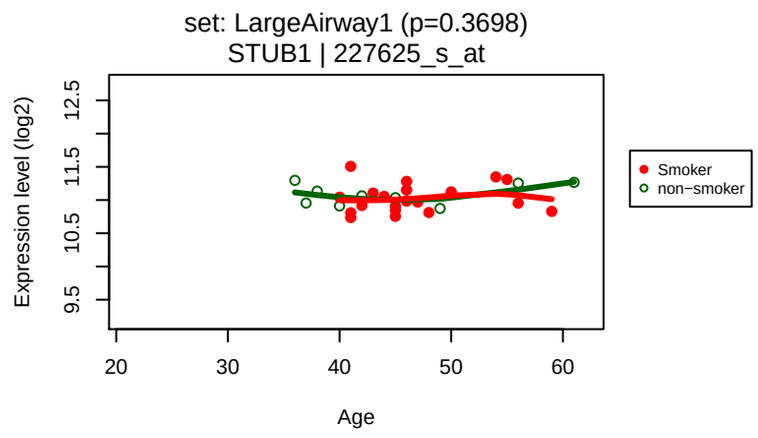

Gene expression of smokers and nonsmokers
gene: STUB1 | 227625_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1008
p=0.0555
p=0.2303
p=0.3698
p=0.0516
p=0.0087
p=0.0839
p=0.0572
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

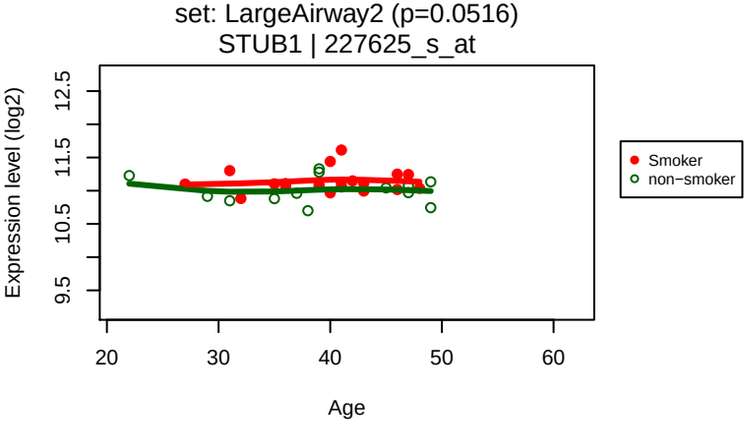

Gene expression ratio (smokers/nonsmokers)
gene: STUB1 | 227625_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1008
p=0.0555
p=0.2303
p=0.3698
p=0.0516
p=0.0087
p=0.0839
p=0.0572
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

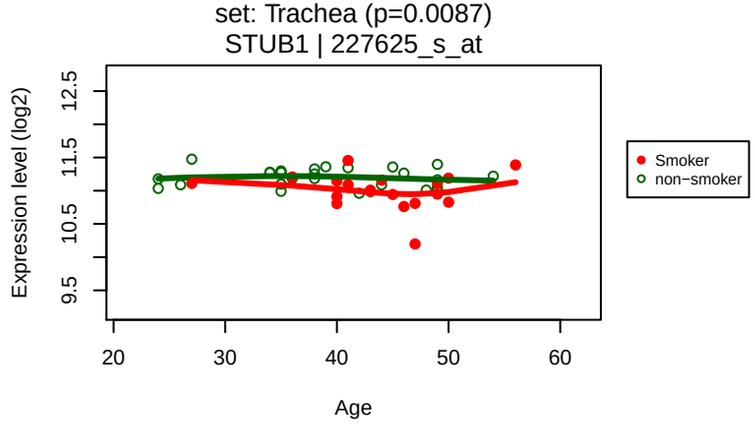

Gene expression ratio in Males
gene: STUB1 | 227625_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0308
p=0.0229
p=0.0578
p=0.0393
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STUB1 | 227625_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6241
p=0.7355
p=0.0048
p=0.9979
SmallAirway2
LargeAirway2
Trachea
Macrophage2

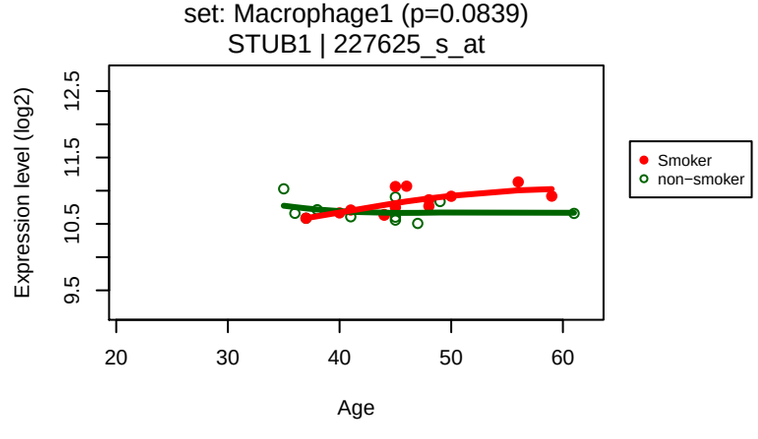

set: SmallAirway1 (p=0.1008)
STUB1 | 227625_s_at
Expression level (log2)
9.5
10.5
11.5
12.5
20
30
40
50
60
Age
Smoker
non−smoker

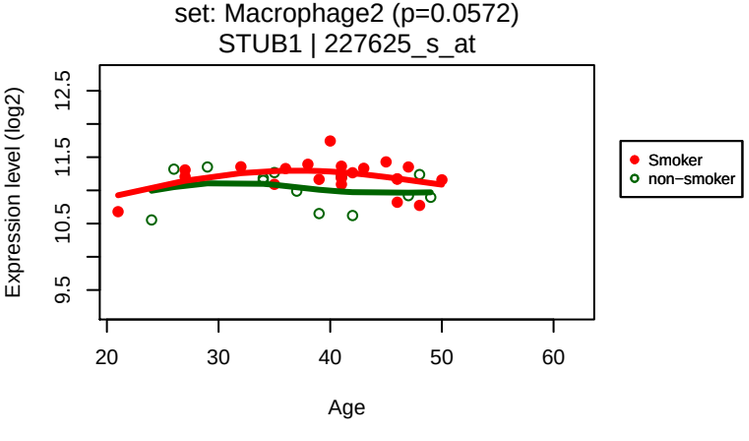

set: SmallAirway2 (p=0.0555)
STUB1 | 227625_s_at
Expression level (log2)
9.5
10.5
11.5
12.5
20
30
40
50
60
Age
Smoker
non−smoker

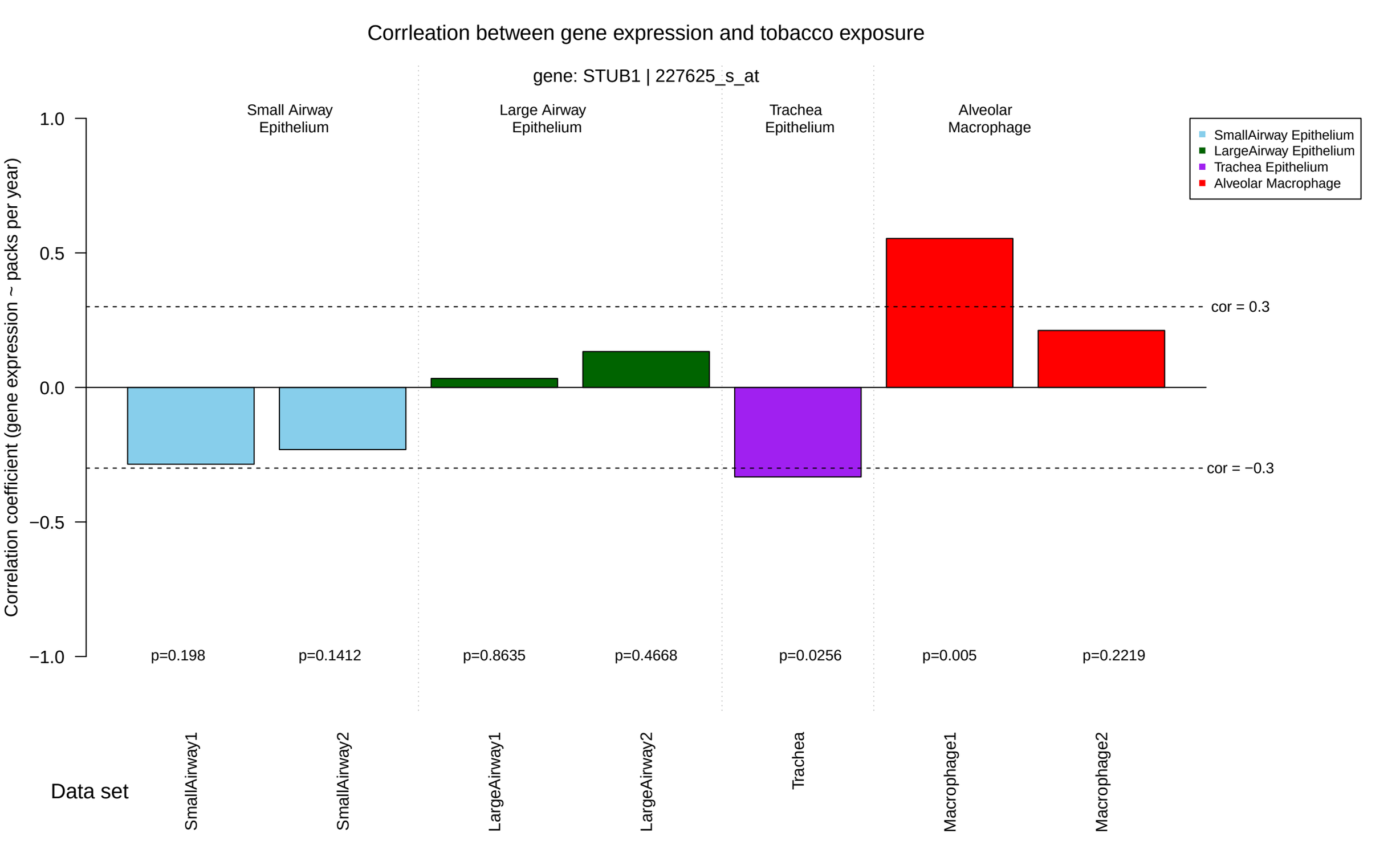

set: LargeAirway1 (p=0.3698)
STUB1 | 227625_s_at
Expression level (log2)
9.5
10.5
11.5
12.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0516)
STUB1 | 227625_s_at
Expression level (log2)
9.5
10.5
11.5
12.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0087)
STUB1 | 227625_s_at
Expression level (log2)
9.5
10.5
11.5
12.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0839)
STUB1 | 227625_s_at
Expression level (log2)
9.5
10.5
11.5
12.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0572)
STUB1 | 227625_s_at
Expression level (log2)
9.5
10.5
11.5
12.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STUB1 | 227625_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.198
p=0.1412
p=0.8635
p=0.4668
p=0.0256
p=0.005
p=0.2219
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2